

Graph-of-Mark: Promote Spatial Reasoning in Multimodal Language Models with Graph-Based Visual Prompting

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GitHub repo
(MIT license),
PyPi library,
and Paper

Spatial Reasoning

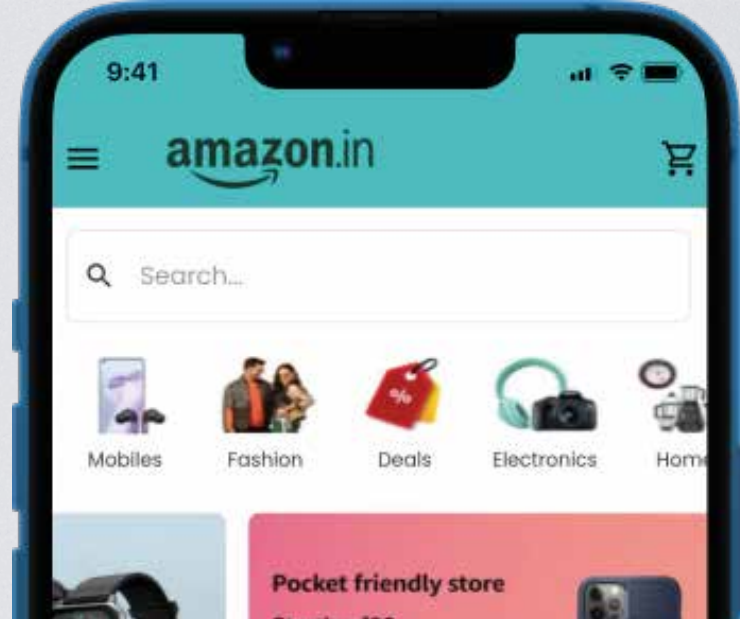
The ability to understand spatial arrangements of objects in 2D and 3D spaces.



Grasp the apple located **in front of the leftmost cup**.



Is the **left kidney below the stomach**?



Type the search bar at the **top of the page**

Harder than you think ...

Surprisingly, state-of-the-art multimodal language models (MLMs) exhibit persistent difficulty in recognizing even basic spatial relationships among two objects only.



Is the car **under the tree**?

Yes



Is the laptop **touching the teddy bear**?

No

GPT-4V



Limitation deeply rooted in their architecture and training objectives, which prioritize **global image-level understanding** over explicit spatial supervision.

Spatial Specialist Models vs. Visual Prompting



Pre-training or fine-tuning MLMs on synthetic data. Multi-task instruction tuning datasets or task-specific datasets derived from real-world or generated images.

Slight architecture edits. E.g., to incorporate new inputs such as depth and scene graphs (SGs).

Requires re-training for new tasks and domains



Modifying the input image pixels before they are fed into the MLM, drawing marks such as colored areas, arrows, boxes, numbers, or alphabets



Set-of-Mark (SoM)
Independent speakable alphanumeric tags

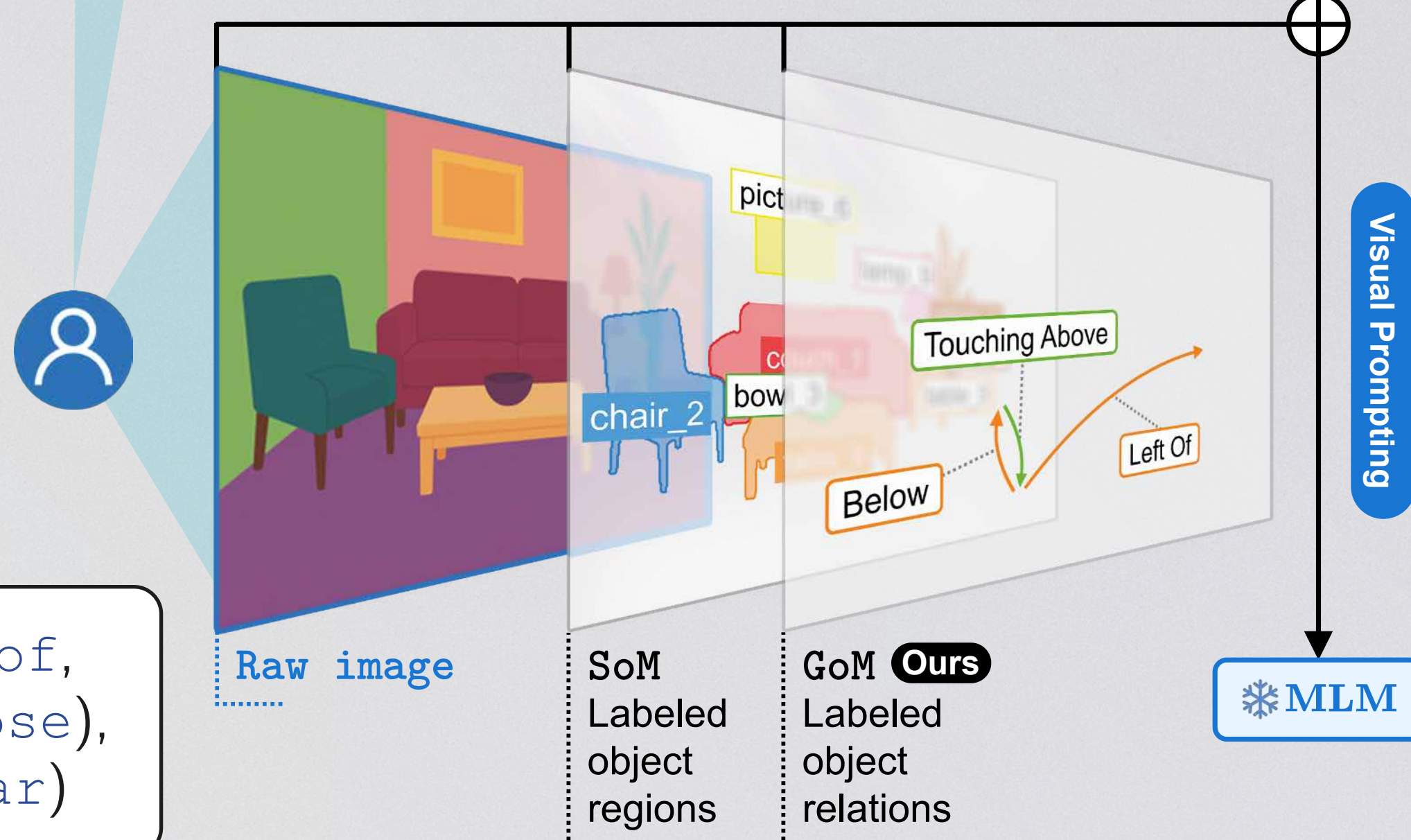
No training or architectural intervention; compatible with any MLM

Ours: Graph-of-Marks (GoM)

- Object detection:** 2D bounding boxes; ensemble of independent detectors and weighted boxes fusion
- Segmentation:** from bounding boxes to region masks
- Relation extraction:** directional (*above*, *below*, *left_of*, *right_of* w/ modifiers *touching*, *close*, *very_close*), depth stacking (*in_front_of*, *behind*), proximity (*near*)

Overlaying scene graphs $P_{MLM}(\cdot | GoM(I), T)$

Which table has a bowl on it?



- Filtering:** (a) retain only objects w/ a class relevant to the query; (b) for each object, retain only the top-k more relevant relations

- SG rendering:** nodes (object areas), edges (arrows); labels; anti-overlapping positioning

Prompting: visual SG, visual and textual SG, <head>- (relation) -><tail>

Experimental Setup

- OWL-V2 () , YOLOv8-X () , Mask R-CNN ()
- SAM-HQ
- MiDaS ()
- WordNet, FastText

Open-source MLMs (, ,) : Qwen-2.5-VL-7B, Gemma-3-4B, LlamaV-o1-11B (*Reasoning*)

2 Tasks (4 Datasets): Visual Question Answering (GQA, VQA_{v1}, VQA_{v2}) Referring Expression Comprehension (RefCOCOg)

Each run (<MLM, prompt method, dataset>) is repeated with 27 combinations of random seed, temperature and top-p

Results (Quantitative, Qualitative, Ablations)

Model	Method	GQA	VQA		RefCOCOg
			v1	v2	
Gemma-3 4B		56.2 $\pm$ 0.11	64.3 $\pm$ 0.14	59.9 $\pm$ 0.19	—
		53.8 $\pm$ 0.34	63.7 $\pm$ 0.15	60.2 $\pm$ 0.37	—
		56.9 $\pm$ 0.34	63.8 $\pm$ 0.07	59.0 $\pm$ 0.18	54.8 $\pm$ 0.91
		58.8 $\pm$ 0.16	65.2 $\pm$ 0.38	62.8 $\pm$ 0.36	56.3 $\pm$ 0.91
		60.3 $\pm$ 0.45	71.5 $\pm$ 0.1	70.2 $\pm$ 0.25	56.4 $\pm$ 0.66
		63.2 $\pm$ 0.07	74.2 $\pm$ 0.16	71.9 $\pm$ 0.01	56.3 $\pm$ 0.58
Qwen-2.5-VL 7B		61.2 $\pm$ 0.28	71.2 $\pm$ 0.07	70.2 $\pm$ 0.15	56.3 $\pm$ 0.42
		61.6 $\pm$ 0.11	77.7 $\pm$ 0.11	73.8 $\pm$ 0.16	—
		53.2 $\pm$ 0.30	65.2 $\pm$ 0.21	67.2 $\pm$ 0.20	—
		61.8 $\pm$ 0.35	65.4 $\pm$ 0.3	68.6 $\pm$ 0.16	55.5 $\pm$ 0.39
		62.5 $\pm$ 0.32	77.9 $\pm$ 0.42	74.0 $\pm$ 0.06	56.5 $\pm$ 0.90
		61.9 $\pm$ 0.41	78.1 $\pm$ 0.37	76.4 $\pm$ 0.21	56.8 $\pm$ 0.74
LlamaV-o1 11B		63.8 $\pm$ 0.13	82.5 $\pm$ 0.48	80.5 $\pm$ 0.03	57.4 $\pm$ 0.21
		60.2 $\pm$ 0.20	81.1 $\pm$ 0.01	80.1 $\pm$ 0.43	57.5 $\pm$ 0.65
		61.9 $\pm$ 0.35	72.6 $\pm$ 0.22	75.3 $\pm$ 0.22	—
		62.0 $\pm$ 0.33	72.8 $\pm$ 0.42	75.5 $\pm$ 0.38	55.3 $\pm$ 1.03
		62.4 $\pm$ 0.3	76.0 $\pm$ 0.49	75.9 $\pm$ 0.23	57.2 $\pm$ 0.79
		62.5 $\pm$ 0.3	77.9 $\pm$ 0.42	80.9 $\pm$ 0.33	57.3 $\pm$ 0.71
GoM (Ours)		67.0 $\pm$ 0.44	79.8 $\pm$ 0.31	83.4 $\pm$ 0.02	56.9 $\pm$ 0.94
		65.0 $\pm$ 0.37	83.2 $\pm$ 0.08	83.6 $\pm$ 0.04	57.6 $\pm$ 0.26
Prompt Legend					
Baselines		Raw image, Segmented objects, Segmented objects + Object Num IDs (SoM)			
		Segmented objects + Relations +			
GoM (Ours)		Object Text IDs /			
		Object Num IDs /			
		Object Text IDs + Relation labels /			
		Object Num IDs + Relation labels			

Comparison between GoM variants and baseline prompt strategies. We report mean $\pm$ standard deviation across decoding runs. Best results highlighted.

GoM incurs very low overhead, averaging 1.13 seconds per image w.r.t. 0.77 and 0.92 for segmentation-only and SoM approaches

GoM demonstrates superior performance across all experimental conditions, showing the efficacy of explicit relational encoding

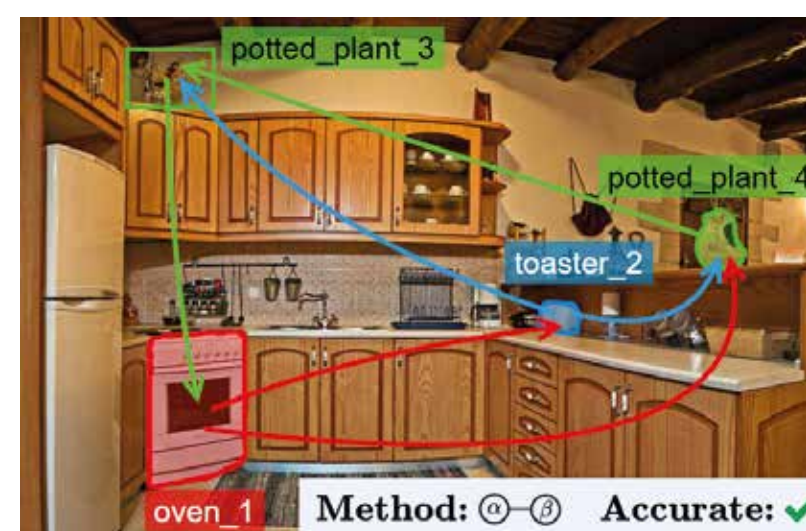
Is the potted plant below the oven?



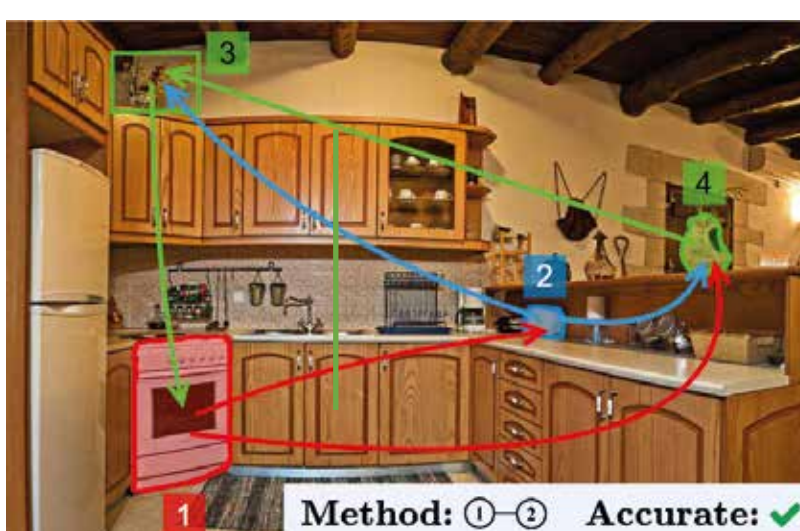
Yes, the potted plant is **below the oven**. It is located **on top of the refrigerator**.



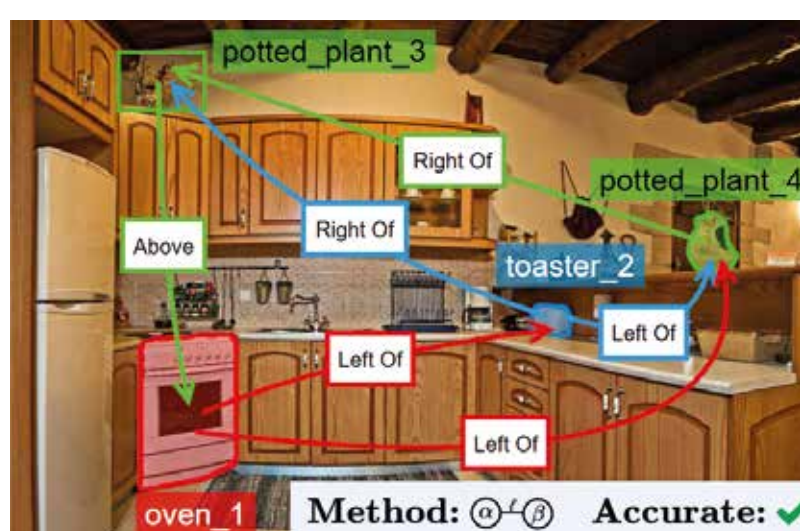
Yes, the potted plant is **below the oven**. According to the image, the potted plant (labeled as number 16) is positioned **below and to the left of the oven**.



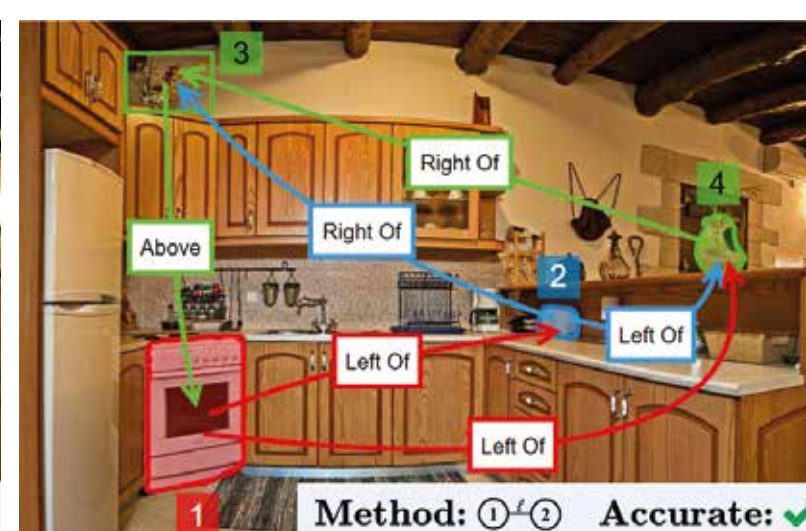
No, the potted plant is not below the oven. The potted plant is located on the countertop to the right of the oven, and it is above the oven in the image.



No. According to the image, the potted plant (labeled as number 4) is located on a countertop to the right of the oven, and it is positioned above the oven when considering the vertical alignment.



No, the potted plant is not below the oven. According to the image, the potted plant (labeled as "potted plant_3") is positioned above the oven (labeled as "oven_1").

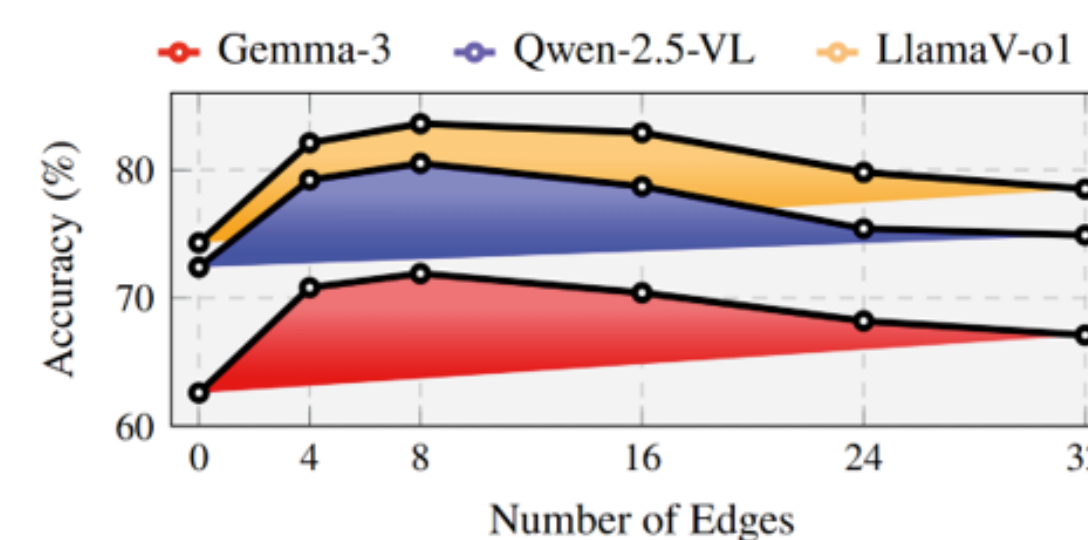


No, the potted plant is not below the oven. According to the image, the potted plant (labeled as 4) is positioned to the right of the oven and above it.

Qwen-2.5-7B on a question from VQA_{v2}

Effect of graph density.

Performance of GoM in VQA_{v2} as a function of #edges in the visual SG. 0 edges = SoM.



Effect of SG modality. Accuracy impact deriving from augmenting the visual (I) and textual prompt (T) with SGs. Gemma-3 results.

